

Exploring Natural Interactions for Implicit User Authentication in VR

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Abstract

We propose an implicit user authentication system for virtual reality (VR), leveraging natural interactions such as grabbing, pointing, and typing. A key challenge in VR authentication is the limited user data available during initial registration, along with the need to pre-collect attacker data for binary-classification settings. In this paper, we address these limitations and develop a practical implicit user authentication solution. We conducted a study with 16 participants, achieving an average Equal Error Rate (EER) of 6.90%, with typing interactions demonstrating a notably low EER of 3.86%. This approach highlights the potential for seamless and unobtrusive authentication in VR environments.

CCS Concepts

• Security and privacy → Authentication; • Human-centered computing → Virtual reality.

Keywords

Behavioral Interaction, Virtual Reality, User Authentication

ACM Reference Format:

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1 Introduction and Related Work

As virtual reality (VR) devices and services become more prevalent, protecting users' personal data and the services and applications associated with these systems has become increasingly important. Unsurprisingly, user authentication is the most fundamental protection technology for these environments [11]. However, traditional authentication methods like PINs and passwords are poorly suited for VR environments, where the absence of physical keyboards and touchscreens makes entering lengthy character sequences cumbersome and degrades the user experience [13, 14]. To address these challenges, a new type of knowledge-based solutions, including spatially aware passwords [4] and 3D object interactions [5], have

been proposed. However, these methods still suffer from a limited key space and cognitive burden, even when combined with behavioral biometrics [8]. Physiological biometric alternatives such as brainwave authentication [1], auditory-pupillary responses [15], and electrical muscle stimulation [3] reduce cognitive burden but require specialized sensors, limiting practical implementation. Recently, behavioral biometric techniques that analyze movement patterns have shown promise for implicit user authentication, utilizing VR peripherals to verify identity in an unobtrusive way.

However, existing behavioral biometric techniques face several practical limitations that hinder their real-world applicability. First, many behavioral authentication systems rely on predefined specific tasks designed solely for authentication purposes. While effective in controlled experiments, this requirement disrupts natural user interactions, making these methods impractical for real-world VR applications where users are immersed in diverse content. Second, these systems often overlook a critical deployment challenge: in practical scenarios, VR devices have limited access to legitimate users' behavioral data for model training, while unauthorized user data (required for robust authentication) is inherently unavailable in situ. Third, VR authentication systems must account for the fact that users often provide limited behavioral data during the initial registration process, creating a data scarcity problem that can compromise model accuracy. Existing research has yet to address these challenges holistically, leaving a significant gap in the field. In detail, previous work [9, 10] overlook these challenges by requiring the user to repeatedly perform a ball throwing task solely for authentication. Targeted to tackle cross-system authentication across separate VR systems, their frameworks lack practicality by obligating the user to provide data from multiple trials across both VR systems in order to perform a single authentication.

To overcome these issues, we propose a novel approach to behavioral biometric authentication in VR environments. Instead of designing dedicated tasks for authentication, we leverage three natural VR interaction tasks—grabbing, pointing, and typing—as means of implicit authentication. These tasks reflect natural user behaviors, ensuring seamless integration into VR content without disrupting immersion. To address the lack of unauthorized user data in real-world deployments, we construct a comprehensive reference dataset of behavioral patterns from unauthorized users beforehand, enabling the authentication model to learn robust decision boundaries. Additionally, we tackle limited legitimate user data during registration by employing feature engineering techniques. We provide a thorough evaluation of extracting meaningful features for VR authentication and using data augmentation techniques, such

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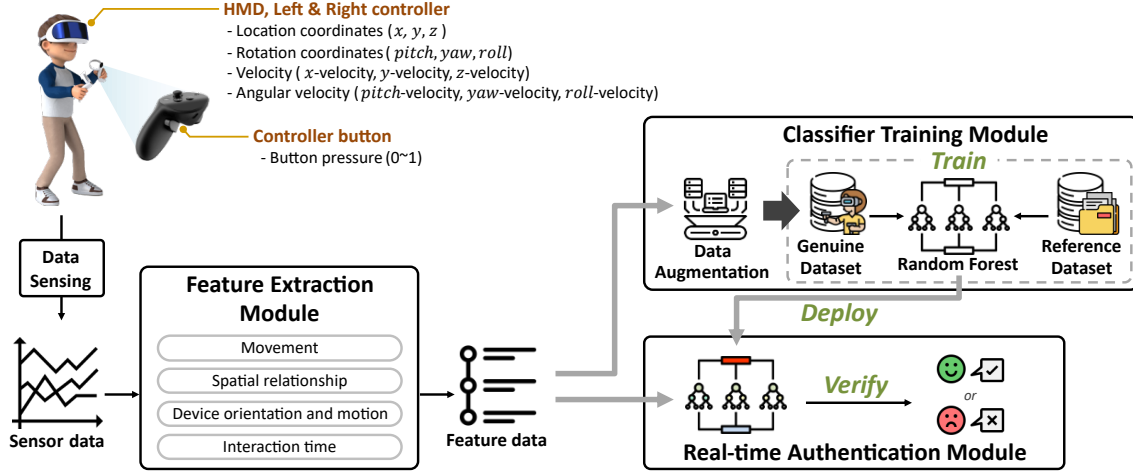


Figure 1: Overview of the proposed authentication system.

as the Synthetic Minority Over-sampling Technique (SMOTE) [2] and the Variational Autoencoder (VAE) [6].

Our proposed approach, leveraging SMOTE for data augmentation, achieved an average Equal Error Rate (EER) of 6.90% across all tasks, effectively addressing the class imbalance in training data and improving authentication accuracy. Notably, for the typing task, our approach achieved an EER as low as 3.86%, demonstrating its capability to capture distinctive and consistent behavioral patterns. Under practical scenarios, our method introduces a minimum of 66% decrease in EER compared to the state-of-the-art method. These results confirm the effectiveness of our approach in providing a practical and robust solution for implicit user authentication in VR environments.

2 Threat Model and Assumptions

Our authentication system aims to differentiate legitimate users from unauthorized users based on behavioral biometrics captured during VR interactions. We assume that unauthorized users may attempt to access the VR system by performing the same standardized tasks as legitimate users. While the unauthorized users have full knowledge of what tasks to perform, they do not have access to the legitimate users' precise behavioral patterns or characteristics that the system uses for authentication. This threat model is based on the assumption that each user exhibits unique and sufficiently consistent behavioral patterns during VR interactions, which can be effectively captured through the proposed feature extraction process. Importantly, the system does not aim to defend against hardware-level attacks or physical compromises of the VR device itself, focusing instead on behavioral differentiation during normal usage scenarios. This model aligns with practical VR usage scenarios, where unauthorized users may obtain physical access to the device and knowledge of the required interactions but must still pass behavioral authentication by naturally performing these tasks. By rigorously evaluating our system's ability to distinguish between users based on their natural interaction patterns, we demonstrate the effectiveness of behavioral authentication in maintaining VR system security.

3 System Design

Figure 1 illustrates the architecture of our proposed behavioral authentication system for VR environments, comprising three interconnected modules: feature extraction, classifier training, and real-time authentication.

3.1 Feature Extraction

The feature extraction module processes continuous streams of raw sensor data from the HMD and controllers to derive behavioral features that characterize user motion patterns. For each interaction task, we extract a total of 224 features across four key behavioral dimensions:

Movement Features. For each device (HMD, LC, RC), we calculate the total movement distance by accumulating the Euclidean distances between consecutive position samples throughout the task execution. This yields 3 features that quantify overall movement intensity.

Spatial Relationship Features. We compute device-to-device distances (HMD-LC, HMD-RC, LC-RC), height measurements from the ground for each device (HMD, LC, RC), and controller distances relative to the user's body center (derived from HMD position). For each of these eight spatial relationships, we extract five statistical measures: maximum, minimum, median, mean, and standard deviation. This comprehensive spatial analysis produces 40 features characterizing the user's spatial interaction patterns.

Device Orientation and Motion Features. To capture detailed motion characteristics, we analyze rotational and velocity data from each device. We record rotation coordinates (pitch, yaw, roll), linear velocity, and angular velocity measurements. The rotation angles undergo sine and cosine transformations before statistical feature computation to ensure a continuous representation of rotational data. We compute the five statistical measures for each transformed motion parameter, generating 180 features that describe fine-grained motion patterns.

Interaction Timing Feature. We measure the cumulative duration of button presses on the dominant-hand controller during task execution to capture the user's temporal interaction patterns.

All feature computations are performed locally on the VR device to maintain data privacy and minimize latency. The extracted features serve as input to both the classifier training process and real-time authentication decisions.

3.2 Classifier Training

The classifier training module builds the authentication model using a random forest classifier, which is well-suited for handling high-dimensional feature data while maintaining computational efficiency for VR systems. The training process utilizes feature data extracted from legitimate users alongside a pre-constructed reference dataset that represents unauthorized user behaviors. This reference dataset is created offline and loaded onto the VR device before deployment. To address the inherent class imbalance where unauthorized samples significantly outnumber legitimate samples, we employ two techniques, SMOTE and VAE, for data augmentation. SMOTE generates synthetic samples of the minority class (legitimate user) by interpolating between existing samples in the feature space. This creates a more balanced training dataset while preserving the distinct characteristics of legitimate user behavior. VAE learns the latent distribution of the legitimate user samples and generates additional synthetic data that captures the variability and nuances of user behavior. Importantly, all training occurs locally on the VR device, maintaining user privacy by avoiding the need to transmit behavioral data to external servers.

3.3 Real-time Authentication

The real-time authentication module operates during VR sessions to verify user identity through behavioral patterns. This module works with the feature extraction pipeline and the trained classifier for ongoing authentication. As sensor data is processed, the module generates feature vectors and evaluates them using the trained random forest classifier. Based on a configurable decision threshold, it outputs authentication scores to accept or reject the current user. This design ensures low computational overhead and seamless integration into VR applications, maintaining an immersive user experience while enhancing security.

4 Evaluation

4.1 User Study

VR Tasks. We evaluate our authentication system using three fundamental VR interaction tasks: grabbing, pointing, and typing. We build upon the framework established by Pfeuffer *et al.* [12]. These tasks were selected as they represent fundamental and natural interactions in VR environments.

The **grabbing** task, illustrated in Figure 2a, simulates direct object manipulation, where users grasp virtual objects using a controller. When a user moves the controller within an object’s boundary and presses the button, the object attaches to the controller and moves in tandem until the button is released. This interaction closely mimics real-world object manipulation, enhancing user immersion.

The **pointing** task, illustrated in Figure 2b, enables distant object interaction through a ray-casting mechanism. Users can manipulate objects by aiming a virtual beam from their controller, entering a hover state over the target object, and pressing the button. This

Table 1: Demographics of user study participants.

Gender			Dominant hand		
Female	8	(50%)	Right-handed	15	(93.75%)
Male	8	(50%)	Left-handed	1	(6.25%)
Age			Heights		
20 - 24	4	(25%)	150 - 159.9 cm	3	(18.75%)
25 - 29	8	(50%)	160 - 169.9 cm	6	(37.5%)
30 - 34	2	(12.5%)	170 - 179.9 cm	3	(18.75%)
35 - 39	2	(12.5%)	180 - 189.9 cm	4	(25%)
40 and more	0	(0%)	Over 190 cm	0	(0%)

interaction extends users’ interaction range while maintaining intuitive control.

The **typing** task, illustrated in Figure 2c, involves text input using a virtual keyboard. Users aim the controller ray at keys and confirm selections by pressing the button. This task is essential for communication, data entry, and command input in VR applications. **Study Design.** To evaluate user performance and collect behavioral data, we designed a study tailored to the specific configurations of these VR interaction tasks mentioned above.

During the **grabbing** task, virtual balls were positioned along the perimeter of a circle with a 0.7 m diameter centered 1.5 m above the ground and 0.5 m away from the user. Balls appear sequentially up to a total of 13 objects per session. Participants conducted 15 sessions for each condition where the virtual ball had a diameter of 0.15 m and 0.3 m.

During the **pointing** task, participants use a ray-casting mechanism to interact with virtual balls positioned along the perimeter of a circle with a 2 m diameter centered 1.5 m above the ground. They align the ray with a ball and press the button to select it. The next ball appears sequentially, up to a total of 13 objects per session. Participants conducted 15 sessions for each condition in which the distance between them and the virtual ball was 2 m and 4 m.

During the **typing** task, participants interact with a virtual keyboard positioned 3.75 m away from them. They select individual keys to input a pre-defined sentence displayed on the screen. Each session required the participant to type one unique sentence and a total of 20 sessions was completed per participant.

Demographics. We recruited 16 participants from a university, and Table 1 summarizes their demographics. Gender was evenly balanced, with 50% male and 50% female. Half of the participants (50%) were aged 25--29, and none over 40 (mean: 28 years, SD: 4.17). Height distribution showed a concentration in the 160--169.9 cm range (37.5%), with smaller proportions in other ranges, and no participants above 190 cm (mean: 169 cm, SD: 9.09). Right-handed participants dominated (94%), with only 6% left-handed.

Collection Protocol. Before the study, we provided participants with a detailed explanation of the study and their rights. After filling the consent form and demographic questionnaire, participants were introduced to how to wear the VR device and our system. All study procedures were conducted in compliance with research ethics. The data collection process was conducted in three separate sessions for each participant. The first two sessions were performed on the same day, with a 30-minute break in between to introduce a

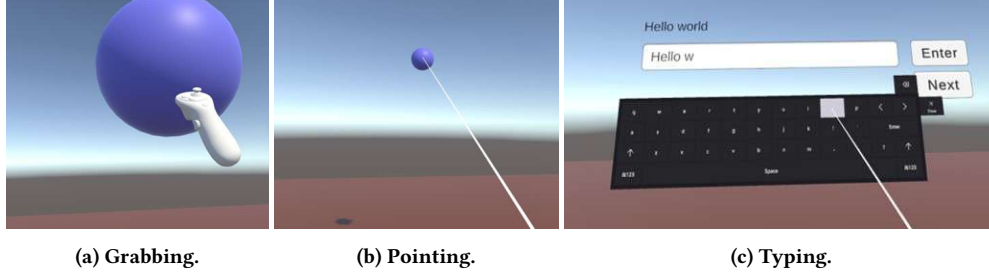


Figure 2: VR interaction tasks implemented in Unity 3D.



Figure 3: User in the study.

short-term temporal gap. Participants removed the VR headset and controllers during the break and reconfigured them before beginning the second study. The third study was conducted at least three days after the second study, introducing a longer-term temporal gap to evaluate the stability of behavioral patterns over time. During each study, participants completed all three VR tasks—grabbing, pointing, and typing—in a fixed order.

Hardware Setup. Participants conducted the study while wearing the Meta Quest Pro, which includes a headset and two controllers, as illustrated in Figure 3. A custom VR application designed to execute the study and collect behavioral data during its execution was implemented using Unity 3D and C#. This application was run on a Windows 10 PC equipped with an Intel Core i9-12900F processor, 32GB of RAM, and an NVIDIA GeForce RTX 3070 GPU. The PC was connected to the Meta Quest Pro headset via a wired connection, ensuring stable execution of the application throughout the study. The sensor embedded in the Meta Quest Pro captured data at a sampling rate of 100 Hz.

4.2 Dataset and Evaluation Framework

The dataset was collected from 16 participants, each completing three study sessions. During each study session, 30 (20)¹ samples were collected per participant for each task, resulting in a total of 90 (60) samples per participant and an overall dataset of 1,440 (960) samples for each task.

To evaluate our system in a manner consistent with real-world VR deployment, we designed an evaluation method that considers the challenge of training authentication models without access to unauthorized user data during deployment. We constructed a reference dataset that represents unauthorized users who are unseen, leveraging all the study data of four randomly selected participants. Performance was evaluated in two scenarios. In **Scenario 1**, the model was trained using the authorized user’s study 1 data and the reference dataset, then tested with study 2 and study 3 data from the authorized user, as well as data from the remaining 11 participants labeled as unauthorized. In **Scenario 2**, the model was trained using study 1 and study 2 data from the authorized user, along with the reference dataset, and tested with study 3 data from both the authorized user and the remaining 11 participants labeled as unauthorized. By including study 2 data in training, Scenario 2

Table 2: EER comparison for three tasks in Scenario 1 (Training with study 1 data and testing with study 2, 3 data).

Task	SMOTE	VAE	No augmentation
Grabbing	13.20 (7.27)	14.82 (9.16)	12.71 (7.34)
Pointing	11.46 (9.32)	12.90 (5.83)	13.71 (8.71)
Typing	8.20 (1.94)	7.12 (2.39)	8.09 (2.16)
Average	10.99	11.61	11.50

* Values in parentheses indicate the median EER.

Table 3: EER comparison for three tasks in Scenario 2 (Training with study 1, 2 data and testing with study 3 data).

Task	SMOTE	VAE	No augmentation
Grabbing	6.19 (2.42)	6.90 (0.76)	6.46 (1.22)
Pointing	10.66 (8.03)	9.65 (3.79)	11.77 (8.19)
Typing	3.86 (0.68)	4.43 (0.45)	4.09 (0)
Average	6.90	6.99	7.44

* Values in parentheses indicate the median EER.

not only provides the model with richer training data but also considers how providing user behavior data over time, albeit at short intervals, affects authentication performance. In real-world applications, VR devices must authenticate users reliably across multiple sessions, where subtle behavioral changes are common. Scenario 2 simulates this challenge by incorporating short-term variability (study 2) into the training process to improve generalization to longer-term changes (study 3).

To address data imbalance in the training data, we used two data augmentation techniques—SMOTE and VAE—and compared their performance with the baseline condition without data augmentation. In addition, to evaluate the effectiveness of each task for implicit user authentication, we conducted experiments respectively for three tasks. For each case, we sequentially label the 12 participants, excluding the four participants used in the reference dataset, as “authorized users” and evaluate the model, providing a thorough assessment of our approach. We evaluate the performance using the EER, which serves as a key indicator of the authentication system’s overall performance.

¹The values in parentheses refer to the typing task.

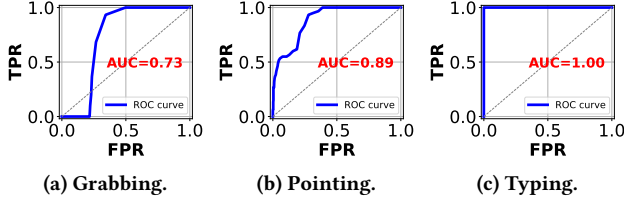


Figure 4: ROC curve of the participant 1 in Scenario 1.

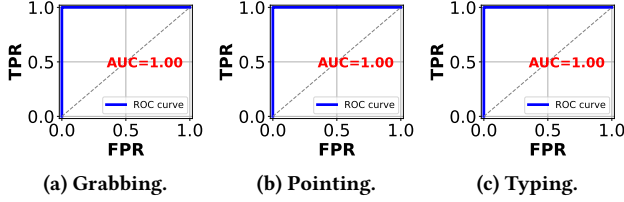


Figure 5: ROC curve of the participant 1 in Scenario 2.

4.3 Experiment Results

Comparison by Varying Data Augmentation. The results in Tables 2 and 3 demonstrate that the use of SMOTE consistently achieved the lowest EER across both scenarios. In Scenario 1, applying SMOTE reduced the average EER to 10.99%, while in Scenario 2, it further improved to 6.90%. This significant reduction underscores SMOTE’s effectiveness in addressing class imbalance during training. VAE also improved performance compared to the baseline, achieving average EERs of 11.61% in Scenario 1 and 6.99% in Scenario 2. However, SMOTE consistently outperformed VAE across all tasks. The baseline configuration without data augmentation resulted in average EERs of 11.50% in Scenario 1 and 7.44% in Scenario 2. While the baseline demonstrated reasonable performance, the results indicate that data augmentation techniques, particularly SMOTE, significantly enhance the system’s robustness.

Comparison across Scenarios. Scenario 2 consistently outperformed Scenario 1; for instance, with SMOTE, it achieved an EER of 6.90% compared to 10.99% in Scenario 1. Figures 4 and 5 illustrate this improvement using the ROC curves of the participant 1, selected to highlight performance differences between scenarios. In Scenario 1, authentication performance varies across tasks, with AUC values ranging from 0.73 (Grabbing) to 1.00 (Typing). In contrast, Scenario 2 achieves perfect discrimination ($AUC = 1.00$) across all tasks. This indicates that utilizing diverse user behavior data over time enhances the model’s capability to adjust to long-term changes for effective user authentication in practice.

Comparison across Tasks. Among the three tasks, typing consistently achieved the lowest EER across all cases, with SMOTE achieving an EER of 3.86% in Scenario 2. This highlights typing as the most effective task for authentication due to its precise and deliberate nature. Additionally, analysis of feature data revealed that the centroid-based Euclidean distance between users was the highest for typing (139.13), compared to pointing (80.22) and grabbing (44.54). This indicates that typing not only yields lower error rates but also exhibits greater inter-user variability, further enhancing its suitability for authentication. In contrast, grabbing and pointing tasks exhibited higher EERs, with grabbing showing slightly better

Table 4: EER comparison across different reference datasets when using SMOTE.

Task	Group1	Group2	Group3	Group4
Grabbing	6.19 (2.42)	6.14 (3.04)	10.71 (7.12)	5.05 (3.79)
Pointing	10.66 (8.03)	15.25 (12.27)	13.56 (3.49)	5.58 (3.79)
Typing	3.86 (0.68)	4.28 (0)	11.86 (7.50)	3.83 (0)
Average	6.90	8.56	12.04	4.82

* Values in parentheses indicate the median EER.

Table 5: EER comparison across different features under the configuration when using SMOTE.

Features	Grabbing	Pointing	Typing	Average
Movement	31.00	32.55	31.57	31.71
Spatial Relationship	20.02	21.09	12.88	18.01
Orientation	18.70	20.48	14.05	17.74
Velocity	17.91	25.24	17.71	20.29
Angular Velocity	16.36	21.93	15.51	17.93
Interaction Timing	37.02	33.10	42.48	37.53

Table 6: EER comparison against state-of-the-art VR authentication method when using SMOTE.

Configuration	Task	Li et al.’s [7]	Ours (w/ SMOTE)
Scenario 1	Grabbing	39.01 (40.65)	13.20 (7.27)
	Pointing	35.35 (30.14)	11.46 (9.32)
	Typing	38.09 (39.30)	8.20 (1.94)
Scenario 2	Grabbing	36.37 (38.49)	6.19 (2.42)
	Pointing	36.41 (39.13)	10.66 (8.03)
	Typing	37.28 (45.10)	3.86 (0.68)

* Values in parentheses indicate the median EER.

performance than pointing in most cases. These results suggest that tasks involving fine-grained, deliberate interactions provide more distinctive behavioral patterns for authentication.

Comparison across Different Reference Datasets. We named the four participants extracted to construct the reference dataset in the previous evaluation as *Group 1*, and partitioned the remaining 12 participants into three non-overlapping groups of four participants each, labeled as *Group 2*, *Group 3*, and *Group 4*. Each group was used as the reference dataset in separate experiments. The results, shown in Table 4, demonstrate that differences in authentication performance emerged depending on the reference dataset. *Group 4* achieved the best average EER of 4.82%, followed by *Group 1* with an average EER of 6.90%. *Group 2* exhibited a slightly higher average EER of 8.56%, while *Group 3* had the highest average EER at 12.04%. Despite *Group 3*’s relatively higher EER, it is important to note that all other groups maintained an EER under 10% in average, which reflects overall satisfactory performance. This suggests that the system’s performance remains robust across different groupings. Moreover, these results highlight the potential for further performance improvement through the refinement of the reference dataset, such as optimizing the selection of participants, which could lead to enhanced authentication model accuracy.

Comparison across Different Features. To check the impact of each feature set on user authentication, we evaluated the performance by training models with the category-specific feature sets divided in 3.1. The orientation, velocity, and angular velocity features are subdivisions of the device orientation and motion features. As shown in Table 5, the best performance was achieved using the orientation features, which resulted in an EER of 17.74%. In contrast, the interaction timing feature, which utilizes only a single feature, resulted in the highest EER of 37.53%. While this performance is relatively lower, it is important to note that the interaction timing feature differs from the others in that it uses only one feature. Despite this, it still contributes to capturing the user's movement patterns to some extent. Interestingly, when all feature sets were used together, the performance was superior to using any individual feature set. This underscores that each feature contributes uniquely to capturing different aspects of user behavior, suggesting that the combined features work synergistically to improve authentication accuracy.

4.4 Comparison with State-of-the-Art

We compared our proposed approach with the state-of-the-art behavioral authentication method by Li *et al.*'s [7] using our collected dataset. As shown in Table 6, our approach significantly outperforms the existing solution across all tasks and scenarios. Specifically, in Scenario 2, the EER was reduced by 83% for grabbing, 71% for pointing, and 90% for typing using our approach, compared to Li *et al.*'s. Our experiments were conducted in a practical VR deployment scenario where it is assumed that authentication model cannot access data from unauthorized users who are unseen during deployment. The existing solution, which relies solely on temporal position coordinates of the user's dominant hand as features for user differentiation, does not consider such real-world constraints, leading to poor performance in our experimental environment. By extracting a diverse set of features that capture user-specific behavior pattern, our approach demonstrate superior authentication performance even in challenging scenario.

5 Discussion

Our findings demonstrate that leveraging natural VR interactions, pre-constructed reference datasets, and data augmentation techniques effectively achieves implicit user authentication in VR environments. This section delves deeper into the implications of our results and identifies areas for improvement and future exploration. **Typing as a Reliable Task.** Among the tasks evaluated, typing exhibited the lowest EER of 3.86% (Scenario 2), significantly outperforming grabbing and pointing tasks. This result highlights the distinctiveness and stability of user behavioral patterns during deliberate and fine-grained interactions, likely due to individual differences in hand movement precision, typing proficiency, or VR adaptation. These characteristics make typing particularly effective for authentication in VR environments. Future studies could explore extending these findings to more varied text input tasks and integrating typing into VR applications like virtual workspaces or secure communications.

Data Augmentation and Diversity. Our results demonstrate that data augmentation techniques, especially SMOTE, are instrumental

in addressing class imbalances. SMOTE achieved a notable performance improvement, reducing the average EER to 6.9% (Scenario 2). While VAE also contributed to performance gains, SMOTE consistently outperformed it across tasks and scenarios. Furthermore, Scenario 2, which incorporated more diverse training data, consistently outperformed Scenario 1 across all tasks. This finding emphasizes the critical role of training data diversity in improving model generalizability. In practical applications, collecting sufficient and representative data during system initialization could significantly enhance authentication accuracy. Additionally, methods to adaptively update models with new user data post-deployment could ensure continued robustness in dynamic real-world settings.

Feature Integration. The integration of multiple feature sets—including movement, spatial relationships, device orientation, and interaction timing—proved superior to relying on any individual feature. Among these, orientation and angular velocity features emerged as particularly effective in capturing unique behavioral patterns, highlighting the value of designing multi-dimensional feature extraction pipelines for robust VR authentication systems.

Limitations. While our study achieved promising results, it has notable limitations. The controlled experimental conditions do not replicate real-world VR complexities, where interactions are fluid and diverse. Additionally, the fixed order of task completion (grabbing, pointing, typing) introduces the potential for order effects, such as fatigue or learning bias, which may influence results. Future work should investigate randomizing task orders to control for such effects. Furthermore, our study does not assess the system's resilience against adversarial attacks. Specifically, our threat model does not account for attackers mimicking victims' behavior using prior knowledge. Future work should evaluate the system's robustness under adversarial scenarios, including shoulder-surfing and imitation attacks, to gain a deeper understanding of its security implications. Lastly, the limited sample size constrains the generalizability of our findings. Expanding participant diversity and enabling real-time model adaptation in the future work would enhance both robustness and applicability across diverse user populations.

6 Conclusion

Our study, conducted with 16 participants, highlights the feasibility of behavioral biometrics for implicit user authentication in VR. It demonstrates the effectiveness of natural interaction tasks—grabbing, pointing, and typing—combined with data augmentation to address critical VR authentication challenges. However, a gap remains in extending these methods from predefined tasks to real-world applications, where users' behavioral interactions are dynamic and varied.

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